Stat123_A4_

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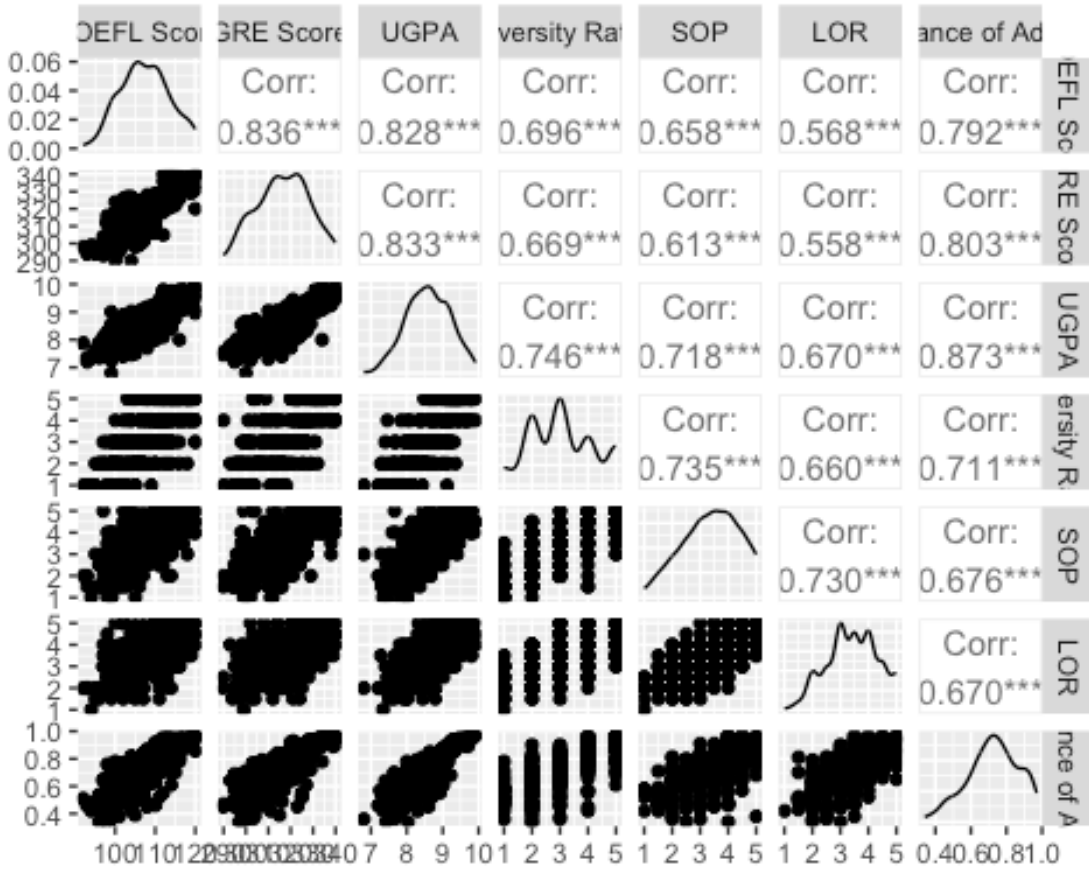
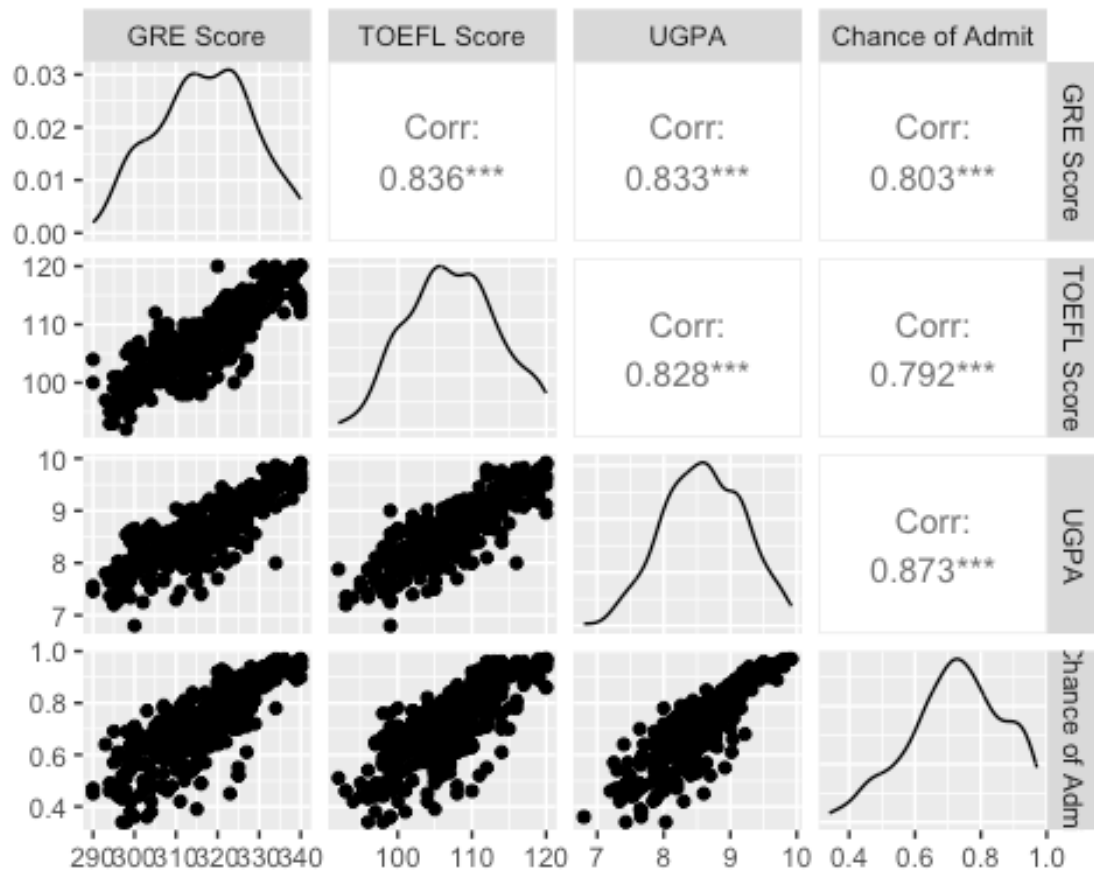
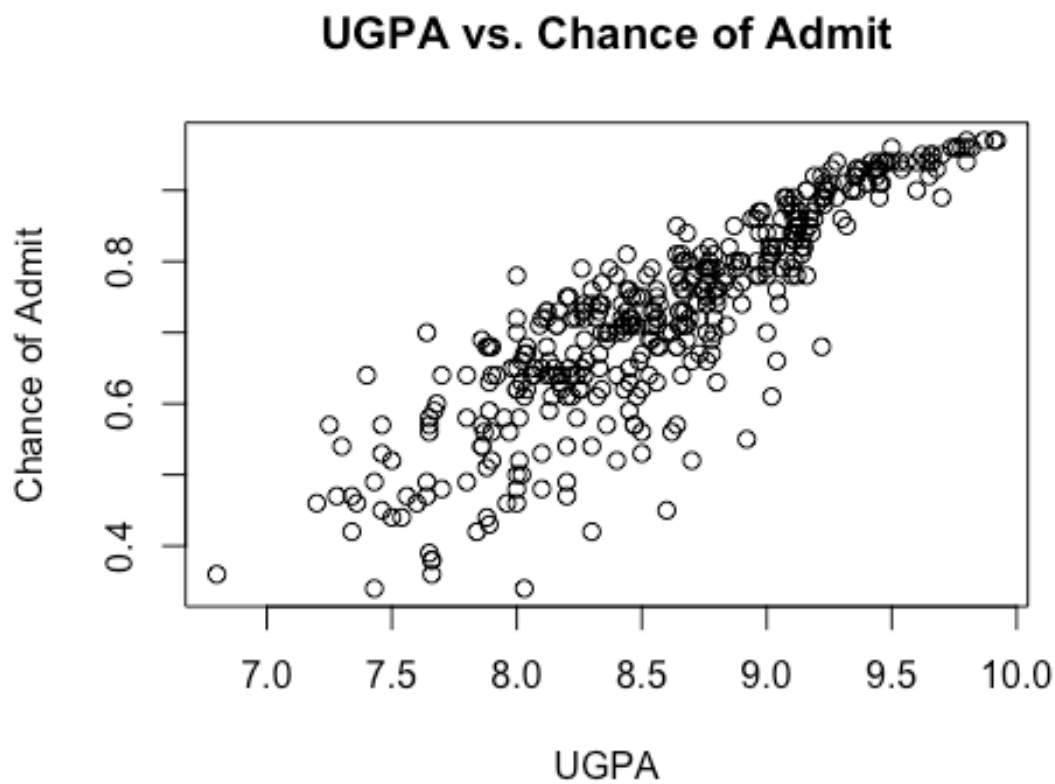
```
library(tidyverse)

Warning: package 'lubridate' was built under R version 4.3.3

── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0
──
✔ dplyr     1.1.4     ✔ readr     2.1.5
✔ forcats   1.0.0     ✔ stringr   1.5.1
✔ ggplot2   3.5.1     ✔ tibble    3.2.1
✔ lubridate 1.9.4     ✔ tidyr     1.3.1
✔ purrr     1.0.2
── Conflicts ────────────────────────────────────────── tidyverse_conflicts()
──
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
conflicts to become errors

library(GGally)

Registered S3 method overwritten by 'GGally':
  method from
  +.gg   ggplot2

#a)
admissions <- read_csv("AdmissionPredict.csv")

Rows: 400 Columns: 8
── Column specification
────────────────────────────────────────────────────────
Delimiter: ","
dbl (8): Serial No., GRE Score, TOEFL Score, University Rating, SOP, LOR,
UG...

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this
message.

ggpairs(admissions |>
  select(`TOEFL Score`, `GRE Score`, UGPA, `University Rating`, SOP, LOR,
`Chance of Admit`))
```




```
#b)
admission <- admissions |> select(`GRE Score`, `TOEFL Score`, UGPA, `Chance
of Admit`)

#c)
ggpairs(admission)
```




```
#the correlation values are generally higher and there is a more clear linear
relationship
#between the data.

#d)
plot(admission$UGPA, admission$`Chance of Admit`,
     main = "UGPA vs. Chance of Admit",
     xlab = "UGPA", ylab = "Chance of Admit")
```




```
cor(admission$UGPA, admission$`Chance of Admit`)

[1] 0.8732891

ugpa_model <- lm(`Chance of Admit` ~ UGPA, data = admission)
summary(ugpa_model)


Call:
lm(formula = `Chance of Admit` ~ UGPA, data = admission)

Residuals:
      Min        1Q    Median        3Q       Max 
-0.274575 -0.030084  0.009443  0.041954  0.180734 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) -1.07151    0.05034  -21.29   <2e-16 ***
UGPA         0.20885    0.00584   35.76   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.06957 on 398 degrees of freedom
```

```
Multiple R-squared:  0.7626,    Adjusted R-squared:  0.762
F-statistic:  1279 on 1 and 398 DF,  p-value: < 2.2e-16
```

```r
#e)
q25 <- quantile(admission$UGPA, 0.25)
q50 <- quantile(admission$UGPA, 0.50)
q75 <- quantile(admission$UGPA, 0.75)

percentiles <- data.frame(UGPA = c(q25, q50, q75))
percentiles
```

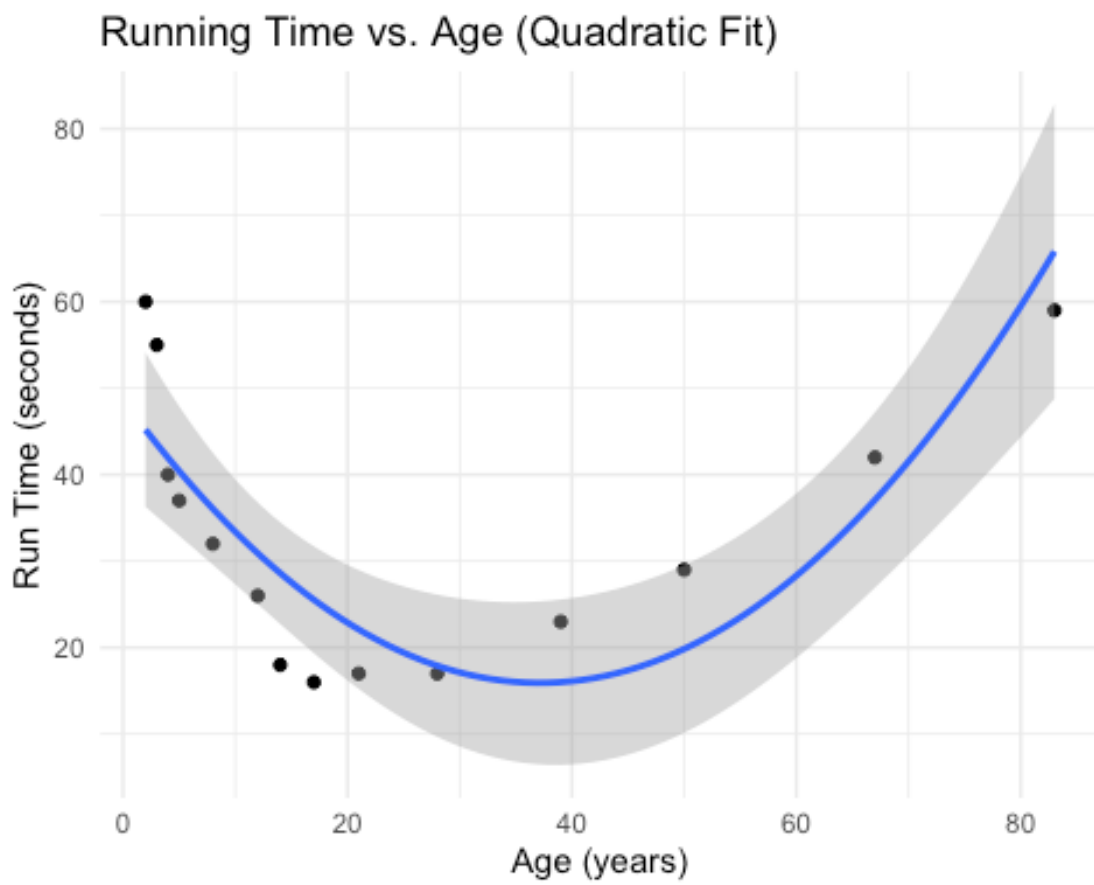
```
      UGPA
25% 8.1700
50% 8.6100
75% 9.0625
```

```r
predict(ugpa_model, newdata = percentiles, interval = "predict", level =
0.88)
```

```
          fit       lwr       upr
25% 0.6347702 0.5261713 0.7433691
50% 0.7266630 0.6181342 0.8351918
75% 0.8211664 0.7125556 0.9297771
```

```r
#f)the residuals may not be evenly distributed across the data by
observation. this could be bad because the model could be less accurate at
lower values and more accurate at higher values etc.
```

```r
#2 a)
age <- c(2,3,4,5,8,12,14,17,21,28,39,50,67,83)
run.time <- c(60,55,40,37,32,26,18,16,17,17,23,29,42,59)

#age is explanatory and runtime is response

#b)
ggplot(data = NULL, aes(x = age, y = run.time)) +
  geom_point() +
  geom_smooth(method = "lm", formula = y ~ poly(x, 2), se = TRUE) +
  labs(x = "Age (years)", y = "Run Time (seconds)",
       title = "Running Time vs. Age (Quadratic Fit)") +
  theme_minimal()
```




```r
#c)
agesq <- age^2
quad_model <- lm(run.time ~ age + agesq)
summary(quad_model)


Call:
lm(formula = run.time ~ age + agesq)
```

```
Residuals:
    Min      1Q  Median      3Q     Max 
-10.617  -5.048  -2.693   6.482  14.793 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 48.635730   4.573375  10.635 3.98e-07 ***
age         -1.762006   0.348789  -5.052 0.000371 ***
agesq        0.023723   0.004306   5.509 0.000184 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.738 on 11 degrees of freedom
Multiple R-squared:  0.7393,	Adjusted R-squared:  0.6919 
F-statistic:  15.6 on 2 and 11 DF,  p-value: 0.000615
```

```r
#d)
predict(quad_model, newdata = data.frame(age = 45, agesq = 45^2), interval =
"predict")
```

```
       fit       lwr      upr
1 17.38403 -4.178831 38.94689
```

```r
#e)
summary(quad_model)$r.squared * 100
```

```
[1] 73.92886
```